

Expanded Linear Responsivity for Earth and Planetary Radiometry

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ABSTRACT: Earth and planetary radiometry requires spectrally dependent observations spanning an expansive range in signal flux due to variability in celestial illumination, spectral albedo, and attenuation. Insufficient dynamic range inhibits contemporaneous measurements of dissimilar signal levels and restricts potential environments, time periods, target types, or spectral ranges that instruments observe. Next-generation (NG) advances in temporal, spectral, and spatial resolution also require further increases in detector sensitivity and dynamic range corresponding to increased sampling rate and decreased field of view (FOV), both of which capture greater intrapixel variability (i.e., variability within the spatial and temporal integration of a pixel observation). Optical detectors typically must support expansive linear radiometric responsivity, while simultaneously enduring the inherent stressors of field, airborne, or satellite deployment. Rationales for significantly improving radiometric observations of nominally dark targets are described herein, along with demonstrations of state-of-the-art (SOTA) capabilities and NG strategies for advancing SOTA. An evaluation of linear dynamic range and efficacy of optical data products is presented based on representative sampling scenarios. Low-illumination (twilight or total lunar eclipse) observations are demonstrated using a SOTA prototype. Finally, a ruggedized and miniaturized commercial-off-the-shelf (COTS) NG capability to obtain absolute radiometric observations spanning an expanded range in target brightness and illumination is presented. The presented NG technology combines a multipixel photon counter (MPPC) with a silicon photodetector (SiPD) to form a dyad optical sensing component supporting expansive dynamic range sensing, i.e., exceeding a nominal 10 decades in usable dynamic range documented for SOTA instruments.

KEYWORDS: In situ atmospheric observations; In situ oceanic observations; Remote sensing; Ocean; Oceanic profilers; Sea/ocean surface

1. Introduction

Passive remote sensing for Earth and planetary science requires observations of signal flux spanning an expanse in brightness, with brightness defined herein as measurable target emission and reflection that increases as flux increases, and darkness herein denotes the opposite. The threshold of signal detectability is determined from the latter by using the noise characteristics of the instrumentation relative to vanishing flux. The environmental range in darkest-to-brightest flux is set, in part, by the spectral characteristics of the target and the illumination, e.g., (spectrally attenuated) flux from the sky, sun, and moon. Insufficient dynamic range in instrument responsivity (i.e., the range in signal over which an instrument produces a characterizable response) inhibits contemporaneous measurements of dissimilar signal levels as follows: Instruments optimized for bright targets may lack sensitivity when pointed at dark targets, and conversely, instruments optimized for dark targets saturate or can be damaged (e.g., unrecoverable hysteresis) when pointed at bright targets. For aquatic Earth science observations, the former is the most prevalent.

For example, instruments designed to observe the bright visible (VIS) spectral domain are typically inadequate for deriving data products from dark spectral domains, e.g., the ultraviolet (UV) and near-infrared (NIR) domains. Similarly, instruments optimized for daytime observations are generally unable to observe nighttime or otherwise low-flux conditions, including extended periods of twilight or near darkness in polar winters. Detector technologies often confer trade-offs in sensing competencies. For example, spectrometers increase spectral resolution compared to radiometers, but often at the expense of dynamic range, signal-to-noise ratio (SNR), stray light rejection, and other required characteristics (Hooker et al. 2012).

Presently, knowledge gaps created by an inability to collect accurate observations across an adequately expansive dynamic range in signal flux span the atmospheric, terrestrial, oceanic, and volcanic disciplines in Earth and planetary science (Hooker et al. 2021a; Bauer et al. 2020). For example, columnar ozone observations using surface instruments cannot operate without sufficient (e.g., solar) illumination for the detector technology (Kyrölä et al. 2006). Aerosol optical depth (AOD) datasets are generally restricted to daytime observations because common photometry instruments lack the dynamic range to perform contemporaneous solar and lunar (or stellar) characterizations (Pérez-Ramírez et al. 2012; Barreto et al. 2019). Measurement of the global and diffuse lunar irradiance using a shadow band radiometer requires an even greater system dynamic range (Witthuhn et al. 2017). Satellite observations for sensors supporting nightly global observing

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are limited in spatial resolution due to flux constraints and are affected by light flicker due to insufficiently rapid aggregation times (Elvidge et al. 2022). Satellite observations of human activities based on nighttime flux levels saturate in bright urban areas in order to observe darker rural areas (Li and Zhou 2017). Although the representative scenarios shown herein primarily correspond to aquatic observations, the opportunities conferred by expanding the dynamic range of observations span multiple disciplines in Earth and planetary sciences and may manifest indirectly, e.g., through advances in spatial or spectral resolution.

The data products from many orbital and suborbital imagers—particularly those intended for both terrestrial and aquatic applications—are inadequate to satisfy community requirements for uncertainties when viewing relatively dark targets, and insufficient radiometric sensitivity limits the maximum observable spatial resolution. For example, low (<20) SNR and nonphysical negative data products are common for shorter wavelengths in the blue to UV domains for airborne hyperspectral imaging of aquatic targets (Kudela et al. 2019, 2024). In addition, spatial resolution or airborne sensing altitude is often limited by field-of-view (FOV) requirements based on flux. Sensitivity limitations at low SNR can also prevent the application of correction schemes for common platform-induced perturbations that degrade signal flux (Zibordi et al. 1999). Technology that expands radiometric dynamic range significantly enhances Earth and planetary science by enabling remote sensing observations spanning seasonal and diurnal differences in illumination conditions, natural variability in target albedo, and an expansive spectral range, the latter of which supports improved bio-optical algorithms (Hooker et al. 2013; Houskeeper 2020).

Expanding the dynamic range of observing technologies is also required to support anticipated advances in spatial and spectral resolution and observation integration times. Increasing spatial and temporal resolution requires observing an increasing range in target brightness because increased resolution captures greater intrapixel variability within remote sensing imagery. For example, a sensor with high spatial resolution viewing an urban target may distinguish dark and bright features (e.g., shadow and concrete) that are aggregated into a single pixel when viewed by a sensor with coarse spatial resolution. In aquatic environments, solar glint from oblique wave facets, which are expressed at a subpixel scale with rapid temporal evolution, can result in aliasing and a false brightening of data products derived from observations of the water surface if the data rate does not resolve spatial or temporal variability in glint to enable removal of glint contaminated observations. The in-water analog of this phenomenon, called wave focusing (WF), can also degrade data products (Zaneveld et al. 2001) if the data rate is not sufficiently rapid to allow an accurate determination of the non-Gaussian central tendency of the light field (Hooker et al. 2020, 2022). Increasing spectral range requires improvements in dark target sensing because atmospheric and aquatic attenuation characteristics spectrally modify signal levels. Recent advances in the dynamic range of above-water aquatic radiometers (Hooker et al. 2018a,b,c) have enabled the spectral expansion of aquatic data products, resulting in improvements to algorithm performance

and target applicability (Hooker et al. 2020; Houskeeper et al. 2021). Presently, the applicable spectral domain for aquatic remote sensing—based on state-of-the-art (SOTA) capabilities of commercial-off-the-shelf (COTS) instrumentation—spans at least 313–1640 nm (Houskeeper and Hooker 2023), although the upper limit of useful spectral range for aquatic applications is unknown.

The subsequent sections document existing requirements for improving the dynamic range of planetary sensors and the technological trajectory associated with satisfying this necessity. First, representative sampling scenarios are provided to demonstrate how expanded dynamic range advances atmospheric and aquatic remote sensing. Next, SOTA capabilities are considered, and opportunities to advance the SOTA are presented in the form of a breadboard—i.e., an experimental arrangement to test feasibility—for the next-generation (NG) instrument design. Three types of instrumentation are organized and discussed as follows: 1) SOTA instruments satisfy the most stringent performance specifications based on established community protocols in terms of data acquisition and the subsequent derivation of data products (sections 2 and 3); 2) legacy instruments have degraded performance specifications with respect to a SOTA alternative with data products satisfying a subset of community protocols and research objectives; and 3) the performance of NG instruments exceeds SOTA specifications and supports the most expansive research requirements (section 4). Legacy and SOTA instruments exist and are widely available from COTS vendors in a multitude of configurations, whereas NG instruments are limited to individual funding successes that move the technology toward exploitable research opportunities.

2. Earth and planetary remote sensing requires an expanded linear dynamic range

Increased radiometric sampling rate (Hooker et al. 2018a,b,c) reveals subpixel variability in observed signals. Aquatic instruments supporting high sampling rates (e.g., 15 or 30 Hz) enable SOTA data processing by resolving intrapixel variability to mitigate bias in above- and in-water data products corresponding to glint contamination (Guild et al. 2020; Houskeeper et al. 2021) and WF effects (Hooker et al. 2022), respectively. Daytime above-water radiometric shipboard observations were obtained at Lake Tahoe, California and Nevada (39.16°N, 120.09°W), on 10 June 2018 under a cloud-free sky by deploying the Compact-Airborne Environmental Radiometers for Oceanography (C-AERO) instrument suite (Hooker et al. 2018c). The data were acquired from the R/V *Hobson's Choice*, a small outboard vessel with a low-height profile to minimize skylight obstruction or adjacency artifacts (Hooker and Zibordi 2005). Lake Tahoe is an oligotrophic freshwater alpine lake, and contemporaneous water sampling indicated that the average concentration of total chlorophyll *a* was 0.061 mg m^{-3} . Wind speed did not exceed 5 m s^{-1} during sampling.

The C-AERO instrument suite includes three radiometers that simultaneously observe the total radiance at the sea surface L_T , the indirect sky radiance L_i , and the global solar irradiance E_s . The radiometers have a consistent architecture

based on silicon photodetector (SiPD) microradiometer technology (Morrow et al. 2010). Each C-AERO radiometer supports 19 wave bands spanning 320–1640 nm (Houskeeper and Hooker 2023), with spectrally and responsivity-dependent predictive dark current (PDC) characterizations (Houskeeper et al. 2021; Hooker et al. 2021b; Guild et al. 2020). The C-AERO radiometers were initially capable of supporting 15-Hz sampling rates, although recent upgrades support up to 30-Hz (Houskeeper et al. 2021; Guild et al. 2020) sampling rates. The radiance radiometers have a full-angle FOV of 2.5° and include integral shrouds to mitigate stray light effects.

Shipboard radiometric observations were configured such that perturbations in data products from the aircraft, sun, and sky were minimized during data acquisition and corrected during data processing. Minimization was achieved by strict adherence to National Aeronautics and Space Administration (NASA) Ocean Optics Protocols (Mueller and Austin 1992) and its subsequent expansions (Mueller et al. 2003), including the use of optical calibration standards traceable to the National Institute of Standards and Technology (NIST). The latter maintained an absolute radiometric capability with uncertainties ($k = 2$ coverage factor) of 2.4%–3.0% for both the radiance and irradiance embodiments of the C-AERO radiometers (Hooker et al. 2018a). A preprocessing correction was applied by using the various available time bases—circular buffers specific to each optical instrument, global positioning system (GPS) records with time, and computer time with timing chip—to convert the near-real-time records on the acquisition computer to more closely match the original real-time sensing of the instruments. The primary correction during the derivation of data products was to remove the brightening in L_T from reflected skylight by using the modeled surface reflectance and surface roughness, with the latter parameterized as a function of the absolute wind speed (Hooker et al. 2002). In addition, the stochastic solar glint spikes in the L_T observations from capillary and surface gravity waves were discretized and rejected from the data, i.e., glint filtering.

Glint filtering—previously applied for shipboard and airborne remote sensing of oligotrophic to eutrophic aquatic targets (Guild et al. 2020; Houskeeper et al. 2021; Houskeeper and Hooker 2023)—removes contamination from glint spikes in L_T observations caused by quasi-randomly oriented wave facets and reveals the radiometrically darker glint troughs, thereby mitigating a bias for deriving aquatic data products (Hooker et al. 2002). Typical high temporal variability in L_T for aquatic observations of a (VIS) peak wavelength (490 nm), $L_T(490)$, is shown in Fig. 1, in which the time series of $L_T(490)$ observations observed at Lake Tahoe is shown for nominal (15 Hz) and reduced (5 and 1 Hz) sampling rates.

Glint troughs are only denoted in Fig. 1 for applicable sampling rates because slower sampling rates do not support adequate discretization of troughs and spikes. For example, glint filtering is ineffective for some legacy sampling rates (e.g., <3 Hz) because the sampling interval is not significantly shorter than the temporal evolution of the wave facets. Raising sampling above 3 Hz not only increasingly mitigates glint contamination (Fig. 2a) but also continues to expand the dynamic (and spectral) range of the observations (Fig. 2b), based on comparing

observations at nominal temporal resolution (15 Hz) with observations resampled at lower temporal resolution (0.1–5 Hz). The 15-Hz sampling rate corresponds to the characteristics of the instrumentation implemented and may not constitute an asymptote.

Besides minimum sampling rate requirements, glint troughs for radiometrically darker wavelengths, e.g., the spectral end members, can be significantly contaminated by noise when using legacy sensing technology with inadequate SNR, dynamic range, or sensitivity. For example, inadequate dynamic range prevents glint filtering for instruments operating at sufficient sampling rates, and inadequate sensitivity can limit the observable spectral range for legacy instruments and those that do not adhere to absolute radiometric scales (Houskeeper and Hooker 2023). Similar limitations may arise for in-water radiometry related to WF effects (Hooker et al. 2022).

Spatial resolution and sampling rate enhancements can improve the discretization of subpixel scale variability but require radiometers to observe increasingly dissimilar targets in terms of radiometric brightness. Operational constraints can further extend dynamic range requirements beyond those arising from target illumination, spectral range, and intrapixel variability. For example, some nadir-pointing or L_T sensors integrated into an aircraft may experience (saturating) high flux when the aircraft is taxiing or transiting over high-albedo surfaces (e.g., concrete and snow or ice, respectively). Although SOTA radiometers built with SiPD microradiometers do not saturate under sunlight—even if pointed at the sun—idling, taxiing, and transiting over high- or low-albedo surfaces can produce significant temperature gradients during field operation, increase dark current variability (which can be mitigated by data processing with PDC), and impose performance limitations on optical instrumentation.

One approach for overcoming detector limitations is to combine two detector technologies wherein one improves upon a limitation of the other. This approach was successfully demonstrated by simultaneously operating an array of SiPD microradiometers and a spectrometer across a similarly expansive spectral range (Hooker et al. 2022). The 10-nm wave bands of the microradiometers provided high accuracy and fast sampling (15 Hz), while the spectrometer provided hyperspectral resolution. This detector configuration was previously defined as *hybridspectral* sensing (Hooker et al. 2012, 2018b, 2022) and is characterized by independent operation wherein the limitations of both detectors are expressed, but contemporaneous observations enable mitigation of separate deficiencies inherent to each detector system. An alternate approach is sensing characterized by dependent operation, in which one detector controls the other such that limitations of both are improved upon and the two detectors function as a single *multidetector*. This detector configuration was previously defined as *hybriddynamic* sensing (Hooker et al. 2018c) and has the advantage that a unified detector capability is created with enhanced performance.

3. Constructed hybriddynamic prototype supports 14 decades in responsivity

An optical detector, including those based on SiPD or photomultiplier tube (PMT) technologies, is the subcomponent of

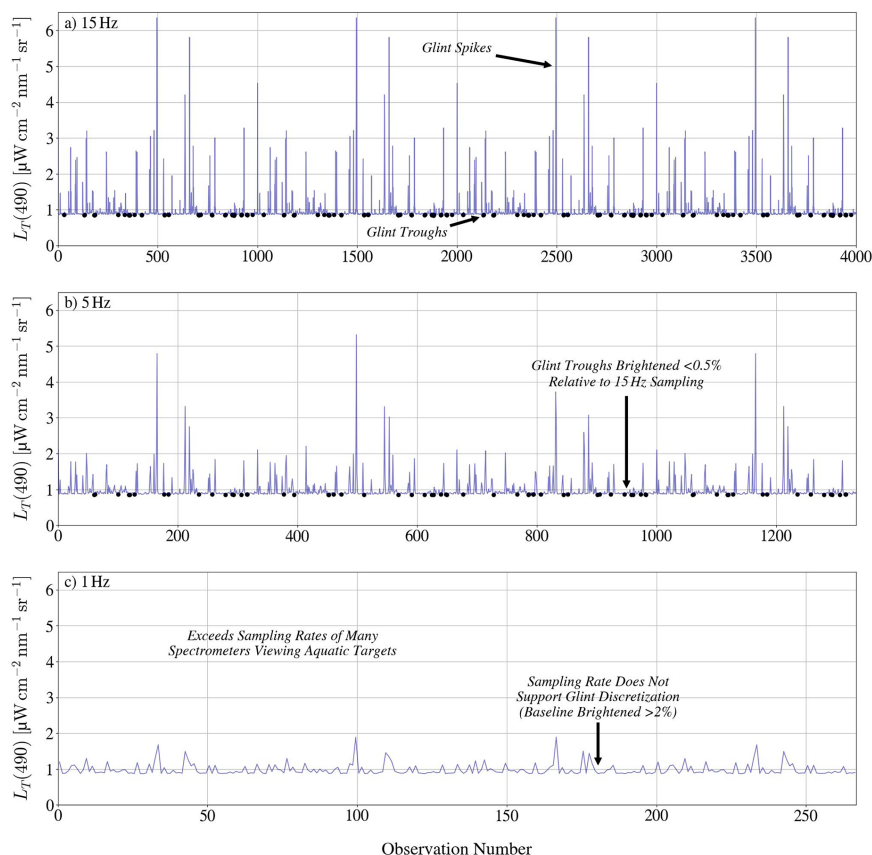


FIG. 1. Retained $L_T(490)$ observations (solid black circles) after glint filtering (Hooker et al. 2002) as a function of temporal resolution as follows: (a) a high (15 Hz) sampling rate (subsequent upgrades to C-AERO support 30 Hz); (b) a moderate (5 Hz) sampling rate representative of legacy SiPD radiometers; and (c) a low (1 Hz) sampling rate representative of, or faster than, legacy spectrometers for aquatic sensing. Characterization of spikes and troughs degrades as the sampling rate decreases. All time series correspond to the same amount of elapsed time, i.e., approximately 267 s.

an instrument system that defines the applicable radiometric dynamic range. Optical detectors for Earth and planetary science typically must support linear radiometric responsivity across an expansive range in flux, while simultaneously enduring the inherent stressors of field, airborne, or satellite deployment. Individual optical detector subcomponents presently limit the dynamic range capabilities of Earth and planetary science radiometers to that of the subcomponent with the most expansive dynamic range but also adequate sensitivity. For example, a PMT provides high sensitivity suitable for observing dark targets but is vulnerable to vibrational and electromagnetic field (EMF) perturbations, and a PMT can be irreversibly damaged from viewing a bright target. SiPD subcomponents provide less sensitivity than PMT subcomponents but are more robust to vibration, EMF, and high signal levels. Common COTS spectrometers improve spectral resolution, but with reduced sensitivity and dynamic range compared to SiPD microradiometers (Hooker et al. 2012).

Multitector designs (i.e., the *hybriddynamic* technology defined in section 2) may satisfy Earth and planetary sensing

requirements, although in some scenarios overall performance might be compromised by vulnerabilities of the individual subcomponents. A multitector embodiment of two separate components is presented herein based on the operational pairing of SiPD and PMT microradiometers optimized for high- and low-illumination conditions, respectively. Overlap in the range of linear responsivity for each separate optical detector component (approximately two decades) ensures a generous flux interval and allows a smooth transition from one detector to the other paired detector. The hybriddynamic SiPD–PMT multitector significantly expands the linear dynamic range for Earth and planetary radiometry, while partially mitigating the significant vulnerability of excess flux exposure for the PMT by shutting down the PMT when SiPD observations exceed a maximum flux threshold. Observations obtained using the SiPD–PMT hybriddynamic technology under lunar illumination are presented in Fig. 3 and provide a spectrally dependent and reproducible opportunity for sensitivity comparisons of detector technologies and hybriddynamic sensing (the brightness of the full moon is approximately 4×10^5 fainter than the sun).

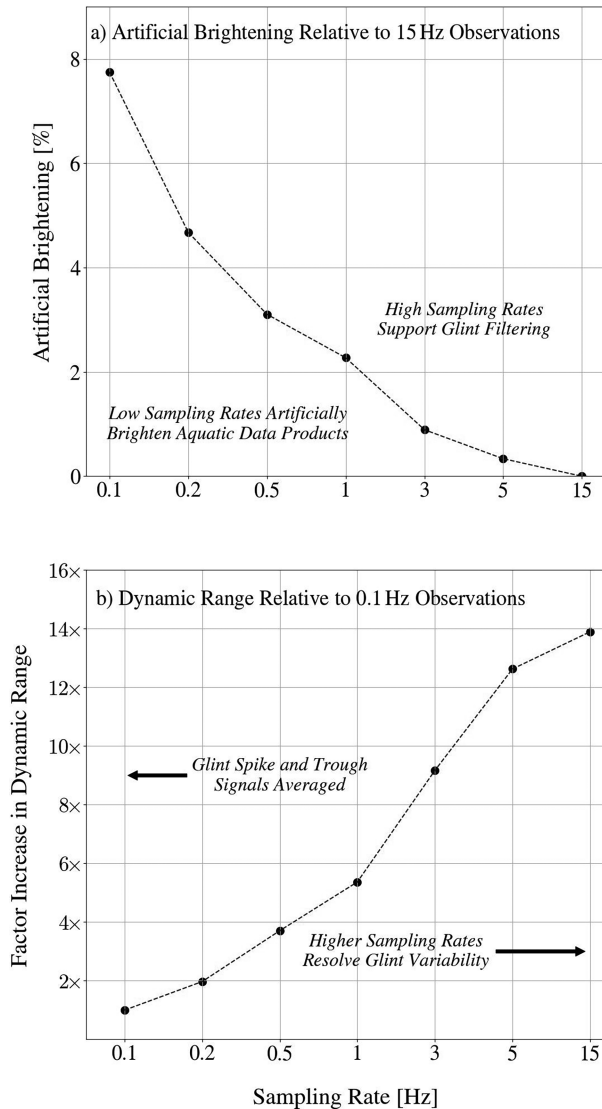


FIG. 2. Comparison of Lake Tahoe $L_T(490)$ observations derived using nominal (15 Hz) and degraded (0.1–5 Hz) sampling rates which are presented using a nonlinear x axis as follows: (a) signal brightening in $L_T(490)$ from aliasing due to inadequate discretization of glint spikes and quantified as the RPD between signals observed after applying a glint filter for observations obtained at sampling rates of 15 Hz or less (15 Hz data are the references in the RPD calculation); and (b) increases in the dynamic range of $L_T(490)$ observations before glint filtering, as a function of increased sampling rate. Observations shown correspond to a wind speed not exceeding 5 m s^{-1} . The glint discretization procedure may be applied for all wind conditions that do not produce whitecaps, i.e., are compliant with the Ocean Optics Protocols. The dynamic range and brightening comparisons are dependent on wind speed and solar geometry, and a representative scenario is shown.

a. Early trajectory supporting expanded dynamic range sensing

A laboratory demonstration test model, i.e., a *breadboard*, was developed in which a PMT was configured as a microradiometer

and architecturally paired with an SiPD microradiometer, wherein the latter controlled the functionality of the former. A subsequent field demonstration model, i.e., a *brassboard*, was constructed in which a radiometer combined SiPD microradiometers with miniaturized and ruggedized PMT microradiometers with compatible form factors and expected survivability typical of the mechanical and environmental stresses associated with oceanic and atmospheric deployments. The brassboard device was called the Ocean Color Underwater Low Light Advanced Radiometer (OCULLAR) and was tested outside the laboratory environment during daytime and nighttime conditions (Hooker et al. 2018c). The OCULLAR radiometer was the first hybrid-namic microradiometer *multitector*. The field trials demonstrated accurate light measurements with high sensitivity obtained in the progression from sunlit to moonlit conditions, thereby creating the opportunity of measuring Earth and planetary targets, including aquatic environments, at night or twilight.

The OCULLAR demonstration project constructed, characterized, and deployed a global irradiance sensor centered at 490 nm. The successful pairing of the PMT and SiPD microradiometers permitted using the latter to determine when light levels were below a safety threshold before applying high voltage (HV) to the PMT, thereby *protecting* the PMT from potentially harmful incident light levels. The large overlap in responsivities between the two detectors (about two decades), and the near-linear response of both, allowed the linear 10 decades of useful dynamic range for an SiPD microradiometer to be extended toward the 14 linear decades for the operationally paired brassboard SiPD–PMT multitector.

Following successful demonstrations of the breadboard and brassboard devices, a hybridnamic prototype instrument was developed and constructed by upgrading the proven architecture of the C-AERO radiometers (section 2), which are part of the Expandable Technologies for Radiometric Applications (XTRA) class of instruments built with 8–19 microradiometers for above- or in-water applications (Morrow et al. 2010). The prototype technology, called the Lowest Observable Light Upgraded XTRA (LOLUX) class of instruments, enhanced the capabilities of the OCULLAR brassboard as follows: 1) The spectral channel count was increased from 2 to 7; 2) the optomechanical design was adjusted for the different lengths of the SiPD microradiometer and the PMT modules to ensure coplanarity between the detectors; and 3) each PMT module was mated to a dedicated base printed circuit board assembly (PCBA) to facilitate a planar connection to purpose-built backplane, acquisition, and aggregator PCBAs, while requiring no hand-wired components.

b. Field demonstration of the LOLUX hybridnamic prototype

A LOLUX irradiance radiometer was deployed on 20 January 2019 approximately 3 m above ground in Anza Borrego State Park (approximately 32.876°N , 116.176°W or 105 km east of San Diego, California) to observe a sunset-to-sunrise time series of $E_\lambda(\lambda)$ —wherein λ denotes wavelength—before, during, and after a total lunar eclipse. The sampling site was selected to minimize light from human activity and was situated near the Great

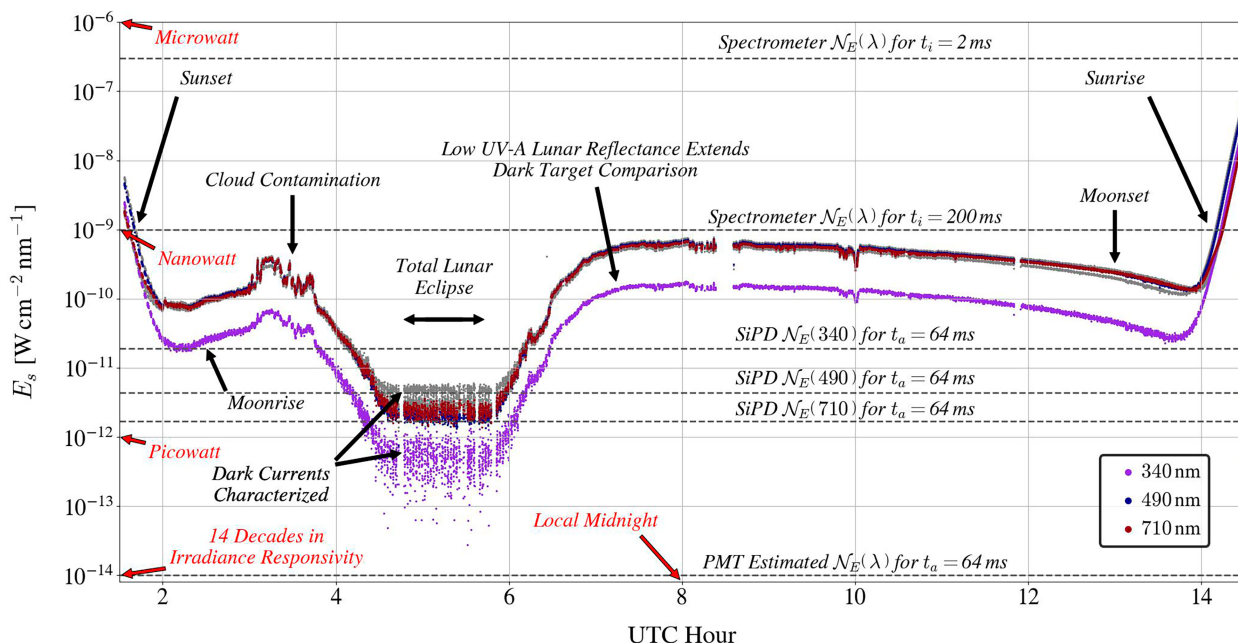


FIG. 3. Nighttime $E_s(\lambda)$ observations encompassing a total lunar eclipse obtained in Southern California. All wave bands are shown in gray, except selected wave bands in the UV (340 nm), VIS (490 nm), and NIR (710 nm) domains are shown in purple, blue, and maroon, respectively. Note that the y-axis scale is presented in watts to preserve notational logic, with microwatt, nanowatt, and picowatt values indicated in red (the scale approaches—but does not reach—femtowatt values). Cloud contamination is primarily responsible for short-term variability, and data gaps correspond to dark current characterizations. Generalized noise equivalent irradiance thresholds are indicated for COTS spectrometers, as well as for microradiometer architectures based on SiPD and PMT optical detector subcomponents.

Southern Overland Stage Route. Post-eclipse full moon observations were obtained under primarily cloud-free conditions, with dark current characterizations performed periodically throughout the time series. Nighttime $E_s(\lambda)$ observations span five decades in dynamic range (Fig. 3), although the prototype was capable of observing higher values during the daytime without resulting in sensor saturation. Assessments of the magnitudes of the derived $E_s(\lambda)$ data products are facilitated, in part, by comparison with the noise equivalent irradiance $\mathcal{N}_E(\lambda)$ defined as the ratio of the standard deviation of the dark signal to the irradiance responsivity. For PMT and spectrometer comparisons, $\mathcal{N}_E(\lambda)$ indicates that the value is generalized across applicable wavelengths. A nominal threshold for defining the validity of noise-limited observations is for signals to exceed $\mathcal{N}_E(\lambda)$ by approximately one and a half orders of magnitude (Hooker et al. 2012).

Published $\mathcal{N}_E(\lambda)$ values for spectrometer and microradiometer instruments are presented with the nominal integration time t_i or acquisition time t_a , respectively, with the distinguishable notations indicating that detector optical subcomponents are dissimilar and warrant consideration when performing direct comparisons. The presentation herein was chosen to maximize consistency in evaluating $\mathcal{N}_E(\lambda)$ across dissimilar technologies, but notable differences in t_i and t_a are as follows: The t_i parameter can be varied during sampling to optimize responsivity for a given flux level, whereas the t_a parameter is fixed by the hardware and firmware architectures; tens to hundreds of spectrometer observations, each individually integrating over a t_i interval, are generally averaged, and data

transmission rates are generally less than 1 Hz; and eight (SiPD or PMT) microradiometer observations, each with t_a of 8 ms, are averaged for data transmission at nominally 15 Hz (i.e., once every 64 ms). Although wavelength dependencies exist for all $\mathcal{N}_E(\lambda)$ values, individual wavelengths are only presented for the SiPD $\mathcal{N}_E(\lambda)$ values because the SiPD values are most similar in magnitude to the darkest observations presented, and therefore, spectral dependencies are highly relevant to the presentation.

Figure 3 shows that even for t_i intervals as long as 200 ms, common COTS spectrometers lack adequate sensitivity to resolve nighttime $E_s(\lambda)$ for the spectral range assessed, whereas SiPD microradiometers provide adequate sensitivity to measure nighttime moonlit $E_s(\lambda)$ for wavelengths longer than the UV (observations during the total lunar eclipse were noise limited for the SiPD microradiometers). Observations in Fig. 3 are atmospheric, and the air–water interface dynamics shown in Figs. 1 and 2 are not applicable. Nonetheless, the expansive range in observations of $E_s(\lambda)$ confirms high dynamic range requirements for advancing atmospheric observations under low-flux illumination (e.g., twilight and nighttime). Sensitivity is challenged in the UV domain, in part, by low flux corresponding to spectral dependencies in the lunar spectrum and atmospheric attenuation. The PMT $\mathcal{N}_E(\lambda)$ values are denoted estimated in Fig. 3 based on dark current observations. PMT microradiometers provide the highest sensitivity and support $E_s(\lambda)$ observations throughout the full sunset-to-sunrise time series (including the total lunar eclipse) but are vulnerable to damage by excessive flux if exposed to daytime flux levels (the

PMT components are powered off when the SiPD components indicate high flux).

Twilight aquatic observations were also obtained using in-water LOLUX instrumentation consisting of a handheld in-water profiler and a surface irradiance instrument deployed on 15 January 2019 from a Coast Guard dock at a marina in Mission Bay, California. Observations were obtained under near-constant illumination and during overcast sky conditions, with a waxing gibbous moon phase (5 days before full) and wind speeds not exceeding 5 m s^{-1} . The marina deployment was an opportunistic proof-of-concept activity, and potential light contamination sources (e.g., harbor lights) were present. Observations of upwelling radiance $L_u(z, \lambda)$ and downward irradiance $E_d(z, \lambda)$ were obtained as a function of water depth z by conducting vertical casts with the in-water profiler. The profiler consisted of upward-pointing irradiance and downward-pointing radiance radiometers mounted on a kite-shaped backplane (Morrow et al. 2010), which added stability to maintain compliant instrument planar geometries. A spectrally identical instrument obtained simultaneous observations to normalize any evolution in $E_s(\lambda)$ during data acquisition. Hydrobaric buoyancy (compressible bladders), in part, supported near-surface loitering, and the vertical sampling resolution (VSR) for the nighttime casts was approximately 4 mm, with planar tilts not exceeding 5° . High VSR supported mitigation of WF effects for deriving surface data products.

Vertical profiles of $E_d(z, \lambda)$ are shown in Fig. 4a. Observations of $E_d(z, \lambda)$ are lower in magnitude than spectrometer $\mathcal{N}_E(\lambda)$ values, consistent with the above-water comparison shown in Fig. 3. Compared to Fig. 3, the differences between the SiPD $\mathcal{N}_E(\lambda)$ values and the $E_d(z, \lambda)$ observations are increased, thereby indicating SiPD microradiometers likely provide adequate sensitivity to observe $E_d(z, \lambda)$ across the full spectral range assessed, although illumination contributions from twilight and marina lights may have been partially responsible for the differences. The PMT microradiometer $\mathcal{N}_E(\lambda)$ values are more than four orders of magnitude darker than the $E_d(z, \lambda)$ observations.

LOLUX observations of $L_u(z, \lambda)$ are shown in Fig. 4b and indicate that nighttime observations of $L_u(z, \lambda)$ are not supported by SiPD microradiometers based on the published values (Hooker et al. 2012) of noise equivalent radiance $\mathcal{N}_L(\lambda)$ except for in portions of the VIS domain with low water attenuation, e.g., 490 nm. Spectrometer observations of nighttime $L_u(z, \lambda)$ are not supported, even for extended t_i values (5000 ms). The PMT $\mathcal{N}_L(\lambda)$ values indicate that the PMT microradiometer subcomponent was adequately sensitive to support the nighttime $L_u(z, \lambda)$ profiles. Suitability of the PMT microradiometer was also confirmed by high coherence of $L_u(z, \lambda)$ with respect to z , with coefficient of determination R^2 values of 0.996, 0.997, and 0.982 for the 340-, 490-, and 710-nm observations, respectively.

LOLUX in-water sampling demonstrated the high efficacy of nighttime aquatic observations. For example, in addition to the high R^2 values for vertical profiles of $E_d(\lambda)$ and $L_u(\lambda)$ data products, derived apparent optical property (AOP) values were similar for daytime and nighttime characterizations. For example, the mean relative percent difference (RPD)

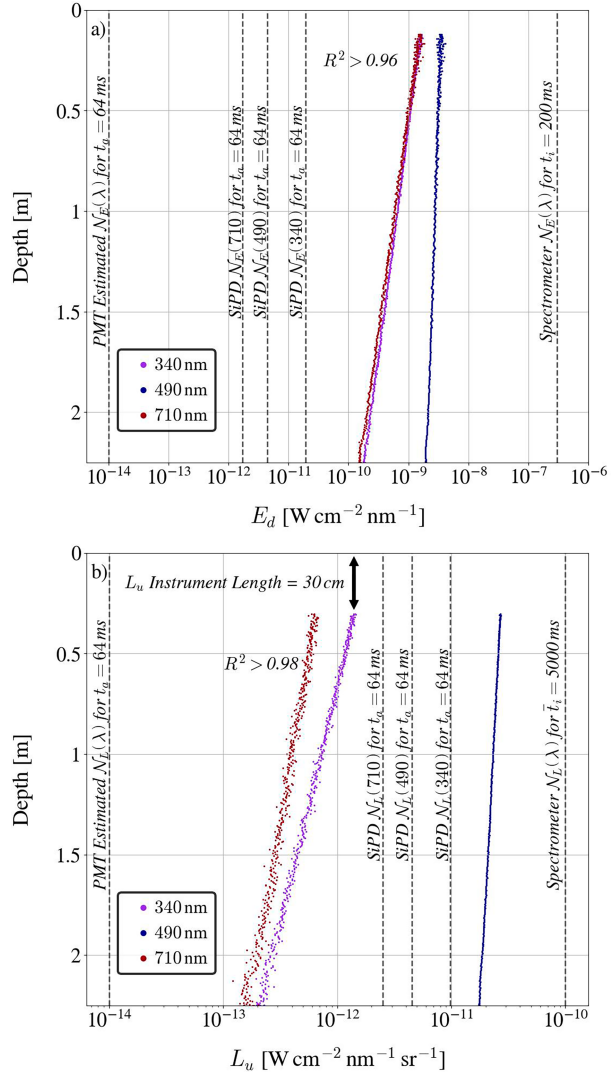


FIG. 4. Nighttime in-water LOLUX optical observations obtained in Mission Bay, California, as follows: (a) $E_d(z, \lambda)$ and (b) $L_u(z, \lambda)$. Wavebands corresponding to UV (340 nm), VIS (490 nm), and NIR (710 nm) domains are shown in purple, blue, and maroon, respectively. Observations are removed for near-surface measurements, wherein the depth was less than the instrument length L_u or was sufficiently close to the air–water interface to cause interface-aperture reflections or other perturbative artifacts.

values derived by comparing daytime and nighttime observations of the spectral diffuse attenuation coefficient $K_d(\lambda)$ and the normalized water-leaving radiance $[L_w(\lambda)]_N$ were -0.9% and -3.8% , respectively (with daytime values used as the reference in the RPD calculations). For comparison, the uncertainty in the absolute calibration was 2.4% – 3.0% (Hooker et al. 2018a).

c. NG opportunities for expanded dynamic range sensing

Although the LOLUX technology concept expands the linear dynamic range, deficiencies are inherent in the multitector prototype design as follows: Redundancies are necessary due

to the requirement for two separate components comprising the optical detectors, each with unique geometric and power requirements and temperature dependencies, all of which result in increased size, weight, power usage, and cost of the multitector system; vulnerabilities to vibrational and EMF perturbations remain due to the PMT component, thereby possibly limiting the system applicability or trajectory to sub-orbital or orbital platforms; spectral range of the system is limited by decreased sensitivity of the PMT component in the NIR spectral domain; the sensitivity of a PMT component may decrease as a function of time; and the procedure for protecting the PMT from excess flux exposure likely does not eliminate vulnerability for targets with rapidly changing or unnatural brightness. These deficiencies are specific to the multitector design but can be significantly mitigated if a single optical component controlling two subcomponents integrates both multitector optical components into a single device. Although the layout of the subcomponents on a printed circuit assembly (PCA) is reminiscent of the existing architecture, an integrated optical component (i.e., closed loop control, electromechanical coupling, and a unified, single optical detector subcomponent) captures the advantages—while avoiding notable deficiencies—of the hybriddynamic multitector concept.

Recent industry advances in solid-state photodetectors constitute an opportunity wherein new subcomponents harmonize the capabilities of separate legacy optical detector components into a single, ruggedized technology. The multipixel photon counter (MPPC) is a silicon photomultiplier that enables the characterization of low-flux conditions by utilizing an array of single-photon avalanche photodiodes, while retaining linearity exceeding five decades of dynamic range, based on a silicon front end that also safeguards against damaging flux conditions (Piatek 2014). MPPC detector components have been incorporated for applications including medical imaging (Kataoka et al. 2015) and light detection and ranging (lidar) for automotive (Nagano et al. 2017) and other industries. Functionally combining two optical detector components (i.e., the SiPD and PMT microradiometers) into a single optical detector component that combines the MPPC and SiPD sensing element subcomponents—hereafter SiMPPC—harmonizes and improves the legacy hybriddynamic multitector. The convergence to a single combined optical component enables architectural optimization to improve the size, weight, complexity, and cost requirements, while removing the PMT microradiometer ruggedizes the system to vulnerabilities associated with the PMT.

Compared to the legacy multitector concept, the SiMPPC system is anticipated to have noteworthy advantages applicable to Earth and planetary science because the integrated solid-state MPPC subcomponent—which has been used in high-risk vibrational environments including autonomous automobiles and drones—is more robust to vibration, EMF, and overflux perturbations, while also supporting observations across an expanded spectral range that includes NIR wavelengths. Compared to available Earth and planetary science radiometers, the SiMPPC system significantly enhances the dynamic range to support Earth and planetary observations of low-illumination environments and dark targets (e.g.,

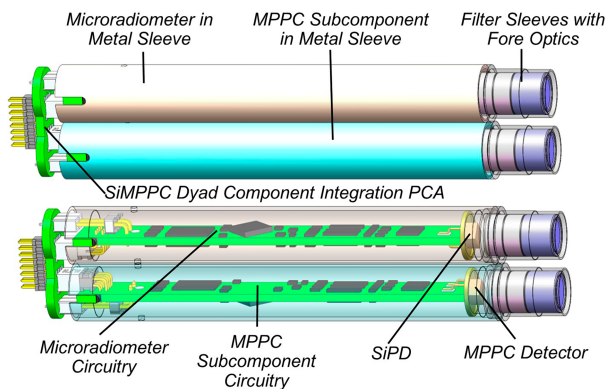


FIG. 5. Notional SiMPPC optical component concept, consisting of a unified dyad of SiPD and MPPC subcomponents, shown in tan and cyan sleeves, respectively. This representation indicates the anticipated manifestation of the optical component based on bread-board evaluations.

spectrally dark data products) over an expansive spectral range. SiMPPC development anticipates and supports NG requirements for increased dynamic range and sensitivity corresponding to future advances in spatial resolution, sampling rate, and spectral range and improves radiometric data quality for airborne simulators or orbital trajectories. One potential NG design concept for implementing SiMPPC technology for Earth and planetary science is described in section 4.

4. Design concept for SiMPPC radiometry

The SiMPPC radiometry concept uses an innovative ultralow-light optical sensing component intended to extend the usable dynamic range and spectral coverage of COTS radiometer systems in relevant operational environments across multiple platforms. The conceptual optical sensing component incorporates a newly developed MPPC subcomponent from Hamamatsu Photonics K.K. (Hamamatsu City, Japan), which uses an array of avalanche photodiode pixels to enable measurements with a dynamic range comparable to that provided by available SOTA subcomponents based on SiPD or PMT technologies. Unlike the latter, the MPPC detector does not have similar limitations in spectral coverage, does not suffer from significant risk factors associated with PMT technologies (such as overexposure) that can be difficult to mitigate, and is not inadequate for significant subsets of the field environments or platforms (e.g., aircraft) used in the community of practice. MPPCs have the potential to overcome these deficiencies and can be combined with an additional qualified sensing subcomponent suitable for higher flux measurements, i.e., the SiPD.

The conceptual SiMPPC optical sensing component utilizes a dyad of two optical detector subcomponents per spectral band: one SiPD and one MPPC (Fig. 5), to provide a dual-stage measurement architecture supporting an expanded linear dynamic range capable of operating in dynamic flux environments where the illumination levels may vary rapidly and unpredictably. The SiPD optical detector sub-

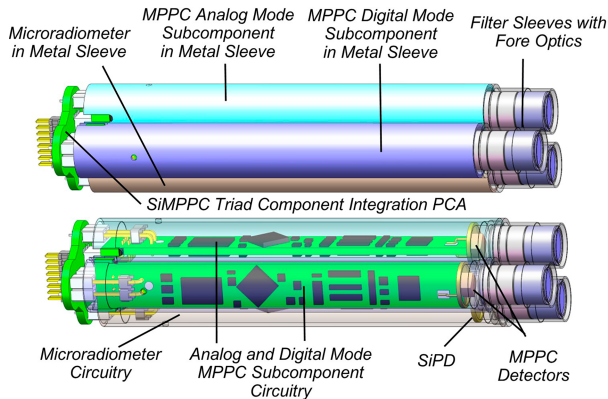


FIG. 6. Notional concept of SiMPPC in a triad configuration with the SiPD (tan) and two MPPC in digital (purple) and analog (cyan) modes. This representation indicates the manifestation of the optical component under a scenario wherein both digital and analog modes are deemed beneficial based on brassboard outcomes.

component contributes 10 decades of linear dynamic range (Hooker et al. 2021b) to the upper end of the component's combined dynamic range. The MPPC optical detector subcomponent may be operated in two separate modes, each adding linear dynamic range, as follows: digital, i.e., photon-counting mode, wherein individual photons create distinct output pulses and trigger a comparator; and analog mode, wherein the output charge from the MPPC element is measured as the total sum of the induced charge from all the photons. Digital mode provides the greatest sensitivity and is useful in ultralow-flux scenarios, whereas analog mode contains greater readout and multiplication noise than digital mode. Analog mode retains linearity at higher flux levels than digital mode because substantial increases in the number of incident photons trigger pulses at frequencies exceeding the ability of the digital mode comparator to distinguish individual pulses.

For an initial SiMPPC dyad design, the MPPC subcomponent operates in analog mode, which ensures dynamic range overlap with the SiPD subcomponent to facilitate calibration and characterization between subcomponents. Operation in analog mode can be tuned to minimize the reduction in sensitivity relative to the digital mode by increasing the time constant of the system amplifier to match the temporal variability of the target signal. If further increases in sensitivity are required, the SiMPPC dyad concept can be upgraded to a triad, consisting of one SiPD and two MPPC subcomponents—one in analog mode and one in digital mode (Fig. 6).

The concept optical block, in which all optical component dyads are mounted, is thermoelectrically cooled to enhance the performance of the SiPD and MPPC subcomponents, but most notably of the latter. The temperature of the optical components is regulated using a thermal component, which consists of a thermoelectric cooling (TEC) subcomponent, combined with a conductive mounting to house the individual optical components, and a heat sink, heat pipe, and fan as necessary. Firmware controls the TEC to support a feedback loop for temperature control.

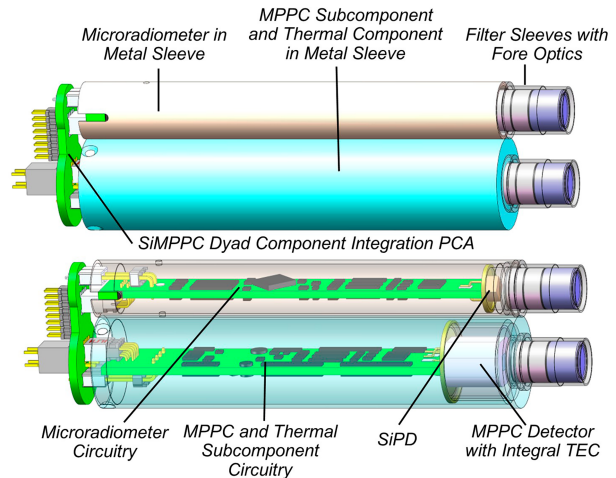


FIG. 7. Notional concept of SiMPPC in a dyad configuration with the SiPD (tan) and the TEC integrated into the MPPC subcomponent (cyan). This representation indicates the manifestation of the optical component under a scenario wherein temperature optimization is deemed beneficial based on brassboard outcomes.

TEC concept options to ensure temperature stability of the MPPC subcomponents force examination of size, weight, channel count, heat-transfer topology, and power requirements. The dyad thermal component described above yields a system with optimized (higher) channel count (greater spectral coverage) and reduced cost and complexity. If the SiMPPC dynamic range decreases due to inadequate cooling performance, however, a heat-transfer topology utilizing dedicated cooling subcomponents per individual MPPC subcomponent creates a more effective temperature control scheme (Fig. 7).

Figure 8 shows a multispectral radiometer system concept leveraging the SiMPPC component as a fundamental building block, with the sensing components clustered in an optimized packaging factor to achieve up to 12 spectral wave bands per system. Each of the 12 clustered SiMPPC optical components has dedicated electronics and embedded processors to implement the configuration and acquisition tasks for the SiPD and MPPC subcomponents. The clustered optical components are situated in the conductive mounting of the thermal component for temperature regulation and interfaced in the instrument to a common control system to provide power, onboard data aggregation, formatting, and telemetry. Under the scenario where SiMPPC dyads are upgraded to triads with digital and analog MPPC modes both included, the system trade-offs are a moderate reduction in system channel count (from 12 to 7), based on a similar packaging factor. Under the scenario where the SiMPPC components are individually cooled, the addition of TEC components to the radiometer system results in a slight reduction in system channel count (from 12 to 10) and greater system cost.

The optimal architecture containing the 12 clustered conceptual optical sensing components supports irradiance or radiance instrument embodiments using a single harmonized electromechanical integration scheme. The irradiance instrument has a proven cosine collector design compatible with the

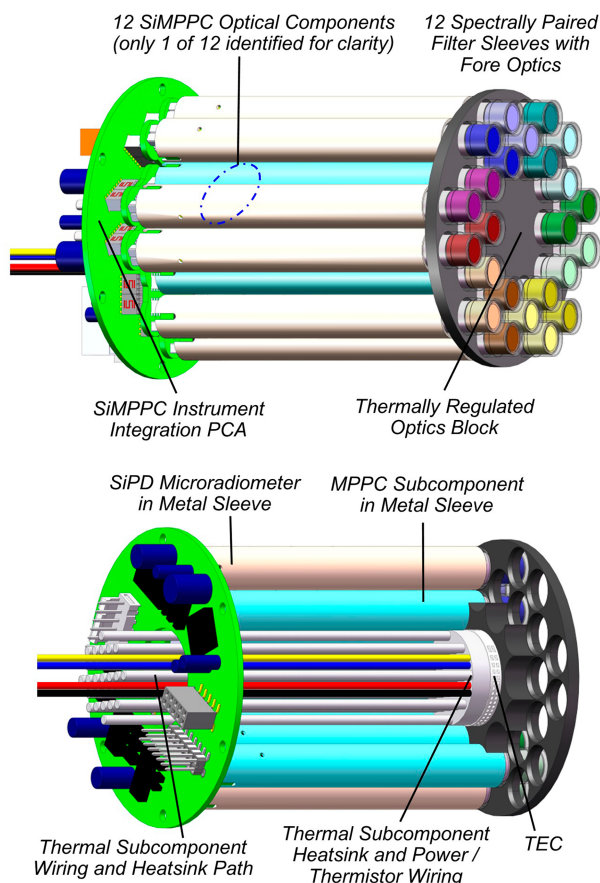


FIG. 8. Notional instrument system concept oriented to show (top) fore optics and (bottom) aft optics, with SiMPPC components strategically oriented to maximize the thermal contact between the TEC and optics blocks. This representation indicates the manifestation of the system concept under a scenario wherein breadboard outcomes are consistent with the breadboard evaluations in Fig. 5 (i.e., dual MPPC modes and further temperature optimization are not deemed necessary in breadboard evaluations).

instrument form factor. The radiance instrument FOV is determined based on the science objectives (i.e., aquatic, terrestrial, or atmospheric), and shrouds are integrated as appropriate. In keeping with established COTS designs, and for the purpose of establishing breadboard dimensions, a 4-in. (101.6 mm) housing diameter is anticipated, with the length determined by measurement geometry, i.e., nominally 16.5 in. (41.9 cm) or less for an irradiance system. The housings are environmentally sealed, purged, and backfilled with dry nitrogen. Each optical instrument system is cabled and interfaced to a data acquisition system directly, or through a control module, depending on the anticipated deployment requirements. For cases wherein multiple instrument systems are required—e.g., an instrument suite comparable to C-AERO supporting L_T , L_i , and E_s pointing geometries—a single control module services a suite of systems to consolidate the data stream to the data acquisition system (Guild et al. 2020).

The SiMPPC radiometry concept is considered NG herein because it advances the SOTA in Earth and planetary science. For example, a nighttime star photometer, the so-called Extinction Camera and Luminance Background Register (EXCALIBUR), consists of a telescope, charge-coupled device (CCD) camera, and filter wheel supporting ultralow-flux observations at seven wavelengths (Pérez-Ramírez et al. 2008b). EXCALIBUR optimizes sensitivity to enable the characterization of nighttime AOD (Pérez-Ramírez et al. 2012) and constitutes one of only a few passive sensing technologies globally with this capability (Barreto et al. 2019). Compared to EXCALIBUR and similar systems, the SiMPPC supports a more expansive dynamic range; i.e., the SiMPPC system exceeds the 10 linear decades of the SiPD system compared to no more than five for EXCALIBUR (Pérez-Ramírez et al. 2008a), which requires pairing with a separate sun photometer system to extend AOD observations through both day and night. The expanded dynamic range supported by the SiMPPC dyad would cover the combined range of daytime and nighttime illumination conditions using a single system with simultaneous detection at a significantly faster sampling of up to 30 Hz (filter wheel devices do not sample simultaneously, which typically results in a net sampling rate significantly less than 1 Hz).

Another comparison is with the RadCam system, which optimizes dynamic range for ocean imaging and is based on complementary metal oxide semiconductor (CMOS) detector technology (Wei et al. 2012). The RadCam system is capable of observing approximately nine decades in dynamic range, in part by varying the sampling rate between approximately 1 and 15 Hz. For an individual observation (i.e., constant sampling rate), the RadCam system supports six decades of dynamic range. Despite dynamic range improvements conferred for ocean observing activities, the RadCam system is not applicable to nighttime or twilight sensing; e.g., the 1-Hz operational mode that targets the lower end of applicable flux conditions is applicable to cloudy daytime observations. Compared to RadCam (nine decades), a system based on the SiMPPC dyad technology supports a more expansive dynamic range without requiring a deleterious decrease in the sampling rate (high sampling rates ensure SOTA data product corrections, including the mitigation of glint and WF effects in aquatic radiometry).

The SiMPPC technology concept also confers advantages in cost and accessibility compared to available—but not comparable—alternatives because SiPD and MPPC technologies are commercially available and the dyad integration is applicable to a COTS—rather than a customized (singular)—design. COTS designs generally reduce system cost, support a broader user base, and achieve greater system continuity because of industry partnership.

5. Conclusions

Remote sensing of dynamic Earth and planetary environments requires new observational capabilities, wherein the spatial and spectral resolution, radiometric quality (including WF or glint mitigation for aquatic applications), and temporal

resolution are significantly advanced beyond ongoing and planned activities (Muller-Karger et al. 2018). A technological trajectory—spanning dynamic range limitations for legacy instrumentation (section 1), an evaluation of capabilities for SOTA COTS (section 2) and prototype (section 3) instruments, and an NG breadboard that advances the capabilities of the prototype instrument (section 4)—is documented herein to support the advancement of imaging capabilities for Earth and planetary science applications.

Advanced data products expand the spectral range of observations and require global—rather than regional—algorithms that are robust to changes in optical complexity (Hooker et al. 2020, 2021a). Recently, algorithms that were applicable to a global range of aquatic ecosystems were demonstrated by utilizing spectrally expansive data products (Hooker et al. 2021a,b, 2020; Houskeeper 2020; Houskeeper et al. 2021, 2022), but deriving spectrally expansive data products is more challenging due to darker aquatic signals. Continuing advances in sampling rate and dynamic range incrementally increase spectral range by revealing nonnegligible flux and exploitable environmental information at sequentially longer (or shorter) spectral end members (Houskeeper and Hooker 2023). Increasing the spectral range of in situ or remote sensing observations requires an increasingly expansive observable dynamic range.

Satellite advances to achieve improved spatial and spectral resolution also require parallel advances in suborbital sensing technology in order to provide validation data products supporting NG satellite missions (Guild et al. 2020). Expanded linear dynamic range for Earth and planetary radiometry supports spectrally expansive airborne remote sensing of the open ocean, coastal zone, and inland waters (Houskeeper et al. 2021) and enables remote sensing under significantly challenging illumination conditions, e.g., dark targets during twilight, nighttime, or polar winter. Expanding the linear dynamic range also supports airborne remote sensing by enabling reductions in the FOV to decrease surface spot size or to maintain surface spot size while increasing airborne sensing altitude, both of which reduce detector flux. An altered FOV can also support system miniaturization by relieving length requirements for Gershun tubes, which benefits the mounting of instrument systems on smaller platforms, e.g., a drone or unmanned aerial vehicle (UAV).

The SiMPPC optical component concept described in section 4 supports Earth and planetary science by increasing the linear dynamic range of radiometers and advancing the capability to observe targets with low signal—both in terms of illumination (e.g., dark targets during twilight, nighttime, and polar winter), spectral domain, or intrapixel variability related to anticipated advances in spatial resolution and sampling rate. The SiMPPC technology concept described herein supports expanded dynamic range compared to spectrometers and SiPD or PMT microradiometers, while miniaturizing and mitigating significant vulnerabilities associated with the LOLUX SiPD–PMT multitector prototype, which had limitations for suborbital or orbital platforms. The improved ruggedness of the SiMPPC system supports an airborne science trajectory for expanded dynamic range Earth and planetary remote sensing—including NG satellite activities—and could

be applicable to satellite missions, e.g., as a miniaturized SmallSat sensor.

Advances in the temporal, spatial, and spectral resolution capabilities of Earth and planetary science radiometers each increase the radiometric dynamic range requirements by revealing intrapixel variability or by darkening the illumination (i.e., aquatic observations at wavelengths highly attenuated by water or FOV reductions). The detrimental limitations associated with low dynamic range are manifested in degraded data products, in which observations are biased, e.g., long integration time observations that are artificially brightened by averaging (non-Gaussian) glint spikes and WF effects, in the case of aquatic above- and in-water radiometry, respectively. Conversely, the benefits of increased dynamic range for Earth and planetary targets have been shown for both legacy and anticipated data products; e.g., intrapixel variability corresponding to glint and WF is adequately discretized. The upper limit in dynamic range for Earth and planetary targets is presently undefined. Continuing advances in spatial and temporal resolution correspond to increasing subpixel variability, and continuing expansions in spectral range reveal measurable and useful signals at nonvisible spectral domains previously below the former limits of detection (Houskeeper and Hooker 2023).

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